

Probability & Statistics

BCA — Semester 3 — 2020 — Answer Sheet

Subject Code: CAST202

Group B

Attempt any SIX questions. [6 × 5 = 30]

4. Calculate correlation coefficient between income and expenditure in foods of certain families of Kathmandu Metropolitan from the following information:

Income (000 Rs) 101 112 131 142 Expenditure in foods (000 Rs) 98 91 211

[5]

Answer: Solution: We calculate Karl Pearson's correlation coefficient.

Income (X) Expenditure (Y) $x = X - \bar{X}$ $y = Y - \bar{Y}$ x^2 y^2 xy 109-2-0.840.641.6 118-1-1.813.241.8 1290-0.800.640
131212.214.842.2 141121.241.442.4 $\Sigma X = 60$ $\Sigma Y = 49$ $\Sigma x^2 = 10$ $\Sigma y^2 = 10.8$ $\Sigma xy = 8$
 $\bar{X} = 60/5 = 12$, $\bar{Y} = 49/5 = 9.8$

$$r = \Sigma xy / \sqrt{(\Sigma x^2 \times \Sigma y^2)} = 8 / \sqrt{(10 \times 10.8)} = 8 / \sqrt{108} = 8 / 10.392 = 0.77$$

Correlation coefficient $r = 0.77$ (strong positive correlation between income and food expenditure).

5. A box contains 50 items of which 20 are defectives. If one item is selected randomly from the box, what is the probability that it is a non-defective item?

[5]

Answer: Solution:

Total items = 50

Defective items = 20

Non-defective items = 50 - 20 = 30

$P(\text{Non-defective}) = \text{Number of non-defective items} / \text{Total items}$

$P(\text{Non-defective}) = 30/50 = 3/5 = 0.6 \text{ or } 60\%$

The probability that a randomly selected item is non-defective is **0.6 (60%)**.

6. What is sampling? The standard deviation of marks in an entrance exam of BCA students is 0.5. How large a sample must be taken in order to be 95% confidence that the error of his/her estimate will not exceed 0.01.

[5]

Answer: Sampling:

Sampling is the process of selecting a subset (sample) from a population to estimate population parameters. It is used when studying the entire population is impractical due to cost, time, or accessibility constraints.

Given:

Population standard deviation (σ) = 0.5

Confidence level = 95% $\rightarrow Z = 1.96$

Margin of error (E) = 0.01

Formula for sample size:

$$n = (Z^2 \times \sigma^2) / E^2$$

$$n = (1.96^2 \times 0.5^2) / 0.01^2$$

$$n = (3.8416 \times 0.25) / 0.0001$$

$$n = 0.9604 / 0.0001$$

$$n = 9604$$

Required sample size = 9604 students.

This is a very large sample because the desired margin of error (0.01) is extremely small relative to the standard deviation (0.5).

7. Calculate the median and mode from following distribution:

Expenditure (000 Rs) 10-20 20-30 30-40 40-50 50-60 60-70 Number of families 7 12 15 13 8 5

[5]

Answer: Solution:

Expenditure Cumulative f
10-20 7 20-30 12 19 30-40 15 34 40-50 13 47 50-60 8 55 60-70 5 60
N = 60

Median:

Median class = class where $N/2 = 30$ th item lies $\rightarrow$ 30-40 class

$L = 30, h = 10, f = 15, cf = 19$

$\text{Median} = L + [(N/2 - cf)/f] \times h$

$\text{Median} = 30 + [(30 - 19)/15] \times 10$

$\text{Median} = 30 + (11/15) \times 10 = 30 + 7.33 = \mathbf{37.33 \text{ (000 Rs)}}$

Mode:

Modal class = class with maximum frequency $\rightarrow$ 30-40 ($f=15$)

$L = 30, h = 10, f_1 = 15, f_0 = 12, f_2 = 13$

$\text{Mode} = L + [(f_1 - f_0)/(2f_1 - f_0 - f_2)] \times h$

$\text{Mode} = 30 + [(15-12)/(30-12-13)] \times 10$

$\text{Mode} = 30 + [3/5] \times 10 = 30 + 6 = \mathbf{36 \text{ (000 Rs)}}$

Median = Rs. 37,330 and Mode = Rs. 36,000

8. A test was given to three candidates taken at random from three provinces of Nepal. The scores of candidates are given below:

Gandaki 97 6 Lumbini 74 5 Bagmati 65 6 Carry out one-way ANOVA.

[5]

Answer: Solution: One-Way ANOVA

Gandaki Lumbini Bagmati 97 6 74 5 65 6 $T_1 = 22$ $T_2 = 16$ $T_3 = 17$

Grand Total $G = 55$, $N = 9$, $k = 3$

Step 1: Correction Factor (CF)

$$CF = G^2/N = 55^2/9 = 3025/9 = 336.11$$

Step 2: Total Sum of Squares (TSS)

$$TSS = (81+49+36+49+16+25+36+25+36) - 336.11 = 353 - 336.11 = 16.89$$

Step 3: Between Groups (SSB)

$$SSB = (T_1^2 + T_2^2 + T_3^2)/n - CF = (484+256+289)/3 - 336.11 \\ = 1029/3 - 336.11 = 343 - 336.11 = 6.89$$

Step 4: Within Groups (SSW)

$$SSW = TSS - SSB = 16.89 - 6.89 = 10.00$$

ANOVA Table:

Source SS df MS F Between 6.89 2 3.445 2.07 Within 10.00 6 1.667 Total 16.89 8

F-critical (2,6) at 5% = 5.14

Since $F = 2.07 < 5.14$, the difference between provinces is **not statistically significant**. We conclude that there is no significant difference in test scores among candidates from different provinces.

Group C

Attempt any TWO questions. [$2 \times 10 = 20$]

9. From the following data, determine average marks of student, standard deviation and coefficient of variation.

Marks0-1010-2020-3030-4040-5050-6060-7070-8080-90No. of people549086586282786670

[10]

Answer: Solution:

Marks	Mid (x)	d=(x-45)/10	fd	fd ²	0-10	545	-4	-216	864	10-20	9015	-3	-270	810	20-30	8625	-2	-172	344					
30-40	5835	-1	-58	58	40-50	6245	0	0	000	50-60	8255	1	8282	282	60-70	7865	2	1563	12					
70-80	6675	3	198	594	80-90	7085	4	280	1120	Total N=646					$\sum fd=0$					$\sum fd^2=4184$				

Assumed mean A = 45, h = 10

Mean: = A + ($\sum fd/N$) × h = 45 + (0/646) × 10 = **45**

Standard Deviation:

$\sigma = h \times \sqrt{[(\sum fd^2/N) - (\sum fd/N)^2]}$
 $\sigma = 10 \times \sqrt{[(4184/646) - 0]}$
 $\sigma = 10 \times \sqrt{6.477} = 10 \times 2.545 = \mathbf{25.45}$

Coefficient of Variation:

CV = (σ/Mean) × 100 = (25.45/45) × 100 = **56.56%**

Results:

- Average marks = **45**
- Standard deviation = **25.45**
- Coefficient of variation = **56.56%**

10. In a normal distribution with mean 200 and standard deviation 20, find the probability that

- a) $P(X > 180)$ b) $P(X < 220)$ c) $P(160 < X < 240)$
d) $P(X > 220)$ e) 10% values are less than what values of X?

[10]

Answer: Solution: Given: $\mu = 200$, $\sigma = 20$

$$Z = (X - \mu) / \sigma$$

a) $P(X > 180)$:

$$Z = (180 - 200) / 20 = -1$$

$$P(X > 180) = P(Z > -1) = P(Z < 1) = \mathbf{0.8413 \text{ or } 84.13\%}$$

b) $P(X < 220)$:

$$Z = (220 - 200) / 20 = 1$$

$$P(X < 220) = P(Z < 1) = \mathbf{0.8413 \text{ or } 84.13\%}$$

c) $P(160 < X < 240)$:

$$Z_{\blacksquare} = (160 - 200) / 20 = -2, \quad Z_{\blacksquare} = (240 - 200) / 20 = 2$$

$$P(-2 < Z < 2) = 0.9772 - 0.0228 = \mathbf{0.9544 \text{ or } 95.44\%}$$

d) $P(X > 220)$:

$$P(X > 220) = 1 - P(X < 220) = 1 - 0.8413 = \mathbf{0.1587 \text{ or } 15.87\%}$$

e) Find X such that $P(X < x) = 10\%$:

From Z-table, $P(Z < -1.28) \approx 0.10$

$$X = \mu + Z \times \sigma = 200 + (-1.28) \times 20 = 200 - 25.6 = \mathbf{174.4}$$

10% of values are less than 174.4.

11. Describe simple random sampling with a suitable example. Write down the process of collecting primary data. Determine First Quartile (Q_1), 7th Decile (D_7) and 80th Percentile (P_{80}) from the following data:

Age in year 10 12 14 16 18 20 22 24 26 No. of people 7 11 21 35 27 17 11 85

[10]

Answer: Simple Random Sampling:

Simple Random Sampling is a probability sampling method where every member of the population has an equal and independent chance of being selected. It is the most basic form of random sampling.

Example: To select 50 students from a college of 1000 students, assign numbers 1-1000 to all students, use a random number table or computer to generate 50 random numbers between 1 and 1000, and select the students with those numbers.

Process of Collecting Primary Data:

1. **Define objectives:** Clearly state what information is needed and why.
2. **Determine population and sample:** Identify who to collect data from and select an appropriate sample.
3. **Choose data collection method:** Direct personal interview, indirect oral investigation, questionnaire, observation, or experimentation.
4. **Design data collection instruments:** Prepare questionnaires, interview schedules, or observation forms with clear, unbiased questions.
5. **Pre-test:** Conduct a pilot study to identify and correct problems.
6. **Collect data:** Administer the instrument to the sample.
7. **Edit and code:** Check for completeness, consistency, and accuracy; convert responses to codes for analysis.

Calculation of Q_1 , D_7 , and P_{80} :

Age (X) f Cumulative f 10 7 12 11 14 18 16 35 18 27 20 11 22 35 24 27 26 11 28 26 51 42

N = 142

First Quartile (Q_1):

Q_1 position = $N/4 = 142/4 = 35.5$ th item
From cf, 35.5th item falls at age **16 years**

7th Decile (D_7):

D_7 position = $7N/10 = 7 \times 142/10 = 99.4$ th item
From cf, 99.4th item falls at age **18 years**

80th Percentile (P_{80}):

P_{80} position = $80N/100 = 80 \times 142/100 = 113.6$ th item
From cf, 113.6th item falls at age **20 years**

Results: $Q_1 = 16$ years, $D_7 = 18$ years, $P_{80} = 20$ years

Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.